RC car — ESP32-S3-CAM brain + RP2040 PWM splitter

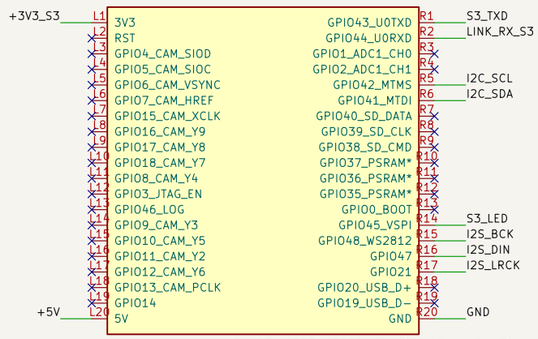
U1 = Freenove L1–L20 / R1–R20

U2 = Pico 1–40 as on board

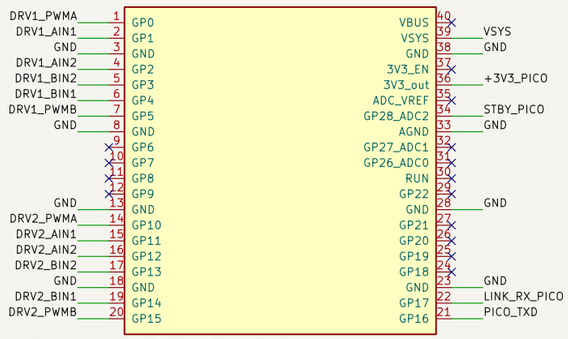
U3/U4 = ROB-14450 JP1-x/JP2-x

MCU BLOCKS

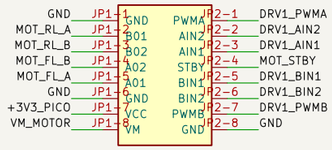
ESP32-S3-CAM (N16R8) — Freenove 2x20 header
U1



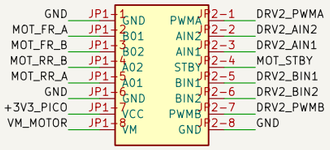
Raspberry Pi Pico (RP2040) — 1–40 header
U2



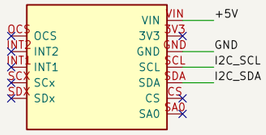
SparkFun ROB-14450 #1 (left: FL=A, RL=B)
U3



SparkFun ROB-14450 #2 (right: RR=A, FR=B)
U4



BMI160 breakout (VIN 3–5 V; own 3V3 reg + 3.9k pull-ups)
U5



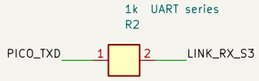
MOTOR DRIVE + IMU

GLUE

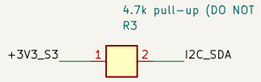
1k UART series
R1



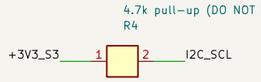
1k UART series
R2



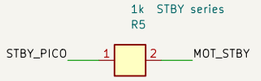
4.7k pull-up (DO NOT FIT)
R3



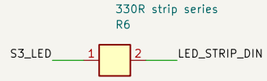
4.7k pull-up (DO NOT FIT)
R4



1k STBY series
R5

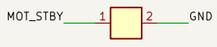


330R strip series
R6

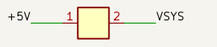


E-STOP / VSYS

E-stop (NC, optional)
SW1

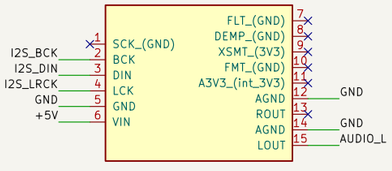


OR link (no Schottky)
R7

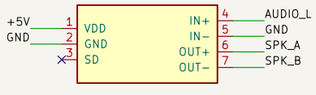


AUDIO + LED

PCM5102A DAC (on-board 3V3 reg, VIN = 5 V)
U6



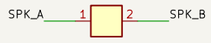
PAM8302A class-D amp
U7



WS2812 strip (5 V)
LED1

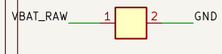


speaker 8R
SP1

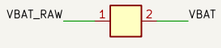


POWER

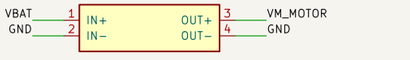
3S LiPo 11.1 V (9.0–12.6 V)
BT1



fuse 7.5 A
F1



XL4015 buck 5 A -> MOTOR RAIL = 8.0 V
PS1

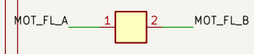


LM2596 buck 3 A -> 5.0 V LOGIC
PS2

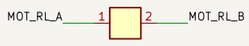


MECHANICAL

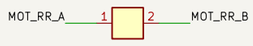
motor FL
M1



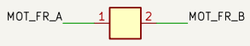
motor RL
M2



motor RR
M3



motor FR
M4



Architecture and rationale: docs/circuit-plan.md
Auto-generated by hardware/gen_schematic.py — do not hand-edit

Sheet: /
File: car.kicad_sch

Title: RC car — ESP32-S3-CAM brain + RP2040 PWM splitter

Size: A3	Date: 2026-10-10	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1